

Literature Status: The Extra Projection Condition in arXiv:2010.05726 Is Unnecessary

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Status: answered in the published revision. The arXiv version imposes an extra condition for strong convergence of the metric projections and says its necessity needs investigation. The 2022 journal version removes that condition completely and proves the unconditional result.

1 Issue in the arXiv version

In arXiv:2010.05726v2, A. B rd llima proves that the iterates $x_n = Tx_{n-1}$ of a nonexpansive, quasi- α -firmly nonexpansive map on a complete CAT(0) space converge weakly to some $x^* \in \text{Fix } T$ [1]. Theorem 3 (arXiv PDF pages 11–12) asserts strong convergence

$$P_{\text{Fix } T}x_n \longrightarrow x^*$$

only under condition (34). The cyclic-projection Theorem 4 on arXiv PDF page 13 similarly requires condition (40). The historical remarks on that page state that the necessity of (40) needs to be investigated.

ArXiv-version issue. *Is the additional condition actually needed to conclude that the metric projections converge strongly to the weak limit?*

2 Published answer

The revised paper appeared as *On a notion of averaged mappings in CAT(0) spaces*, *Functional Analysis and Its Applications* **56** (2022), 27–36 [2]. Its Theorem 3 (Russian original, journal pages 46–47; PDF pages 9–10) states the same general convergence theorem with no extra condition:

Published answer. *If T is nonexpansive and quasi- α -firmly nonexpansive on a complete CAT(0) space, then $x_n = Tx_{n-1}$ converges weakly to some $x^* \in \text{Fix } T$, and $P_{\text{Fix } T}x_n$ converges strongly to x^* .*

Theorem 4 (journal pages 47–48) gives the corresponding unconditional statement for cyclic projections, and Theorem 5 (pages 48–49) does so for averaged projections.

3 Why the condition disappears

The published proof supplies a short completion of the arXiv argument. The following formulation isolates the decisive step. Put $C = \text{Fix } T$ and $p_n = P_Cx_n$. Since (x_n) is Fej r monotone with

respect to C , for $m \geq n$ the projection inequality gives

$$d(p_m, p_n)^2 \leq d(x_n, p_n)^2 - d(x_m, p_m)^2.$$

The nonincreasing sequence $d(x_n, C)$ has a limit, so (p_n) is Cauchy and $p_n \rightarrow p \in C$.

Because $x^* \in C$, the same projection inequality yields

$$d(x_n, p_n)^2 + d(p_n, x^*)^2 \leq d(x_n, x^*)^2.$$

Also $|d(x_n, p) - d(x_n, p_n)| \leq d(p, p_n) \rightarrow 0$. Hence

$$\liminf_n d(x_n, p) \leq \liminf_n d(x_n, x^*).$$

But bounded weak convergence in a complete CAT(0) space satisfies the Opial inequality: if $p \neq x^*$, then the reverse strict inequality holds. Thus $p = x^*$, proving $P_C x_n \rightarrow x^*$ without any additional regularity condition.

This also shows directly why the conditional arXiv proof was only one step short: its Cauchy argument already produced p ; the Opial property identifies that limit with x^* .

4 Provenance

A direct attack rediscovered both a geodesic estimate and the Opial completion before the published-version check. The exact theorem is, however, already present in the author's revised publication, so no new mathematical result is claimed here. The broad sentence in the arXiv paper leaving a possible metric notion of monotone operators for future research is not a precise theorem-sized target and is not promoted.

References

- [1] A. B rd llima, *On a notion of averaged operators in CAT(0) spaces*, arXiv:2010.05726v2 (2020).
- [2] A. B rd llima, *On a notion of averaged mappings in CAT(0) spaces*, Funct. Anal. Appl. **56** (2022), 27–36; Russian original: Funktsional. Anal. i Prilozhen. **56**:1 (2022), 37–50; DOI: 10.1134/S0016266322010038 and 10.4213/faa3859.